



Your Thesis Title Here

A thesis submitted in partial fulfillment of
requirements for the award of the degree of

Master of Science (MS)

by

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(12MS345)

5th Year BS-MS

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July 25, 2026

Declaration

I, Your Name, roll number 12MS345, a student of the Department of Physical Sciences of the 5th Year BS-MS program of Indian Institute of Science Education and Research Kolkata, India, hereby declare that this thesis is my own work and, to the best of my knowledge, it neither contains materials previously published or written by any other person, nor has it been submitted for any degree/diploma or any other academic award anywhere before. I have used the originality checking service to prevent inappropriate copying.

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This is to certify that the thesis titled “Your Thesis Title Here” submitted by Your Name, a student of the Department of Physical Sciences of the 5th Year BS-MS program of Indian Institute of Science Education and Research Kolkata, India, is based upon his own research work under our supervision. We also certify, to the best of our knowledge, that neither the thesis nor any part of it has been submitted for any degree/diploma or any other academic award anywhere before.

.....

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Prof. Supervisor Name

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Date: July 25, 2026

Place: IISER Kolkata, Mohanpur

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Affiliation

Date: July 25, 2026

Place: IISER Kolkata, Mohanpur

Acknowledgements

Let's just say, I've never been the best at predicting how much effort something will require...

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List of Symbols and Abbreviations

$|\Psi\rangle$ Quantum state vector

c Speed of light in a vacuum

Todo list

 This is a margin note. It automatically fits the page margin.	4
 This is an inline todo note, useful for longer comments or reminders inside the text body.	4

Abstract

This thesis explores a novel approach to constructing a finite-dimensional holographic scalar field theory.

1

Getting Started

Welcome to the IISER Kolkata MS and PhD thesis template. This document serves as the official documentation, demonstrating how to use the template effectively to produce a beautiful, correctly formatted thesis.

1.1 Project Structure

The template is divided into several modular components to keep your workspace clean:

- `main.tex`: The root document where you configure metadata and include other files.
- `IISERKthesis.cls`: The core LaTeX class file that manages page layout, margins, headers, and frontmatter.
- `IISERKphysics.sty`: A custom package that loads mathematics tools, defines theorem environments, and configures code blocks.
- `frontmatter/`: Contains individual files for your abstract, acknowledgements, certificate, and declaration.
- `chapters/` and `appendices/`: Directories to house your main textual content.

1.2 Metadata Configuration

In `main.tex`, you will find several macros to define your personal details. These automatically populate the title page, declaration, and certificate:

```
1 \title{Your Thesis Title Here}
2 \author{Your Name}
3 \rollno{12MS345}
4 \program{5th Year BS-MS}
5 \department{Department of Physical Sciences}
6 \supervisor{Prof. Supervisor Name}
```

1.3 Compilation

This template relies on **biber** for bibliography management and **minted** for syntax-highlighted code blocks. Because of **minted**, you *must* compile with the `-shell-escape` flag.

The easiest way to compile is using the included `.latexmkrc` configuration:

```
1 | latexmk main.tex
```

This automates multiple compilation passes and stores all auxiliary files (like `.aux` and `.log`) neatly in a hidden `.build_tex/` folder.

1.4 Class Options and Toggles

The template automatically generates several lists at the beginning of the document (Table of Contents, List of Figures, List of Tables, List of Listings, Nomenclature, and a Todo list).

You can easily disable any of these by passing toggles to the document class in `main.tex`:

```
1 | % Disables the Todo list and List of Tables
2 | \documentclass[notodo, nolot]{IISERKthesis}
```

Available toggles include:

- **notodo**: Completely disables the `todonotes` package and removes the Todo List. Ensure you use this flag for your final submission!
- **nolof**: Removes the List of Figures.
- **nolot**: Removes the List of Tables.
- **nolol**: Removes the List of Listings.
- **nonom**: Removes the Nomenclature (List of Symbols).

2

Features and Customization

This chapter highlights the advanced typographical features and custom environments built into the `IISERKphysics.sty` package.

2.1 Font Customization

You can easily switch out the document fonts by using the font macros in `main.tex`. These macros take an optional LaTeX package name and a styling command.

```
1 % Example: Use 'tgheros' for sans-serif headings, 'tgtermes' for Times main text
2 \TitleFont[tgheros]{\sffamily}
3 \TextFont[tgtermes]{\rmfamily}
4 \MathFont{newtxmath}
```

If you leave these commented out, the template elegantly defaults to Garamond.

2.2 Theorem Environments

We provide beautifully styled theorem environments powered by `tcolorbox`. These boxes are fully integrated with `cleveref` for smart cross-referencing.

Theorem 2.1 (No-Hair Theorem)

Stationary black holes are characterized entirely by mass M , charge Q , and angular momentum J .

Lemma 2.2

This is a lemma without a title. Notice how the numbering follows the theorem counter chapter-wise.

Definition 2.3 (Event Horizon)

The boundary defining the region of space from which nothing can escape.

Remark 2.4 (Historical Note)

The term was popularized by John Wheeler in 1967.

You can reference these dynamically using `\cref{}`: [theorem 2.1](#) and [definition 2.3](#).

2.3 Mathematics and Physics

Essential math packages like `amsmath`, `mathtools`, `slashed`, and `tensor` are pre-loaded. The `physics` package provides excellent shorthand for bra-ket notation and derivatives:

$$\hat{H} |\Psi\rangle = i\hbar \frac{\partial}{\partial t} |\Psi\rangle \quad (2.1)$$

You can also log symbols to the Nomenclature automatically. For instance, consider the speed of light c .

2.4 Code Listings

The template utilizes `minted` for professional code blocks. Crucially, we use the `accsupp` package to render line numbers uncopyable, so readers can effortlessly copy your code from the PDF without grabbing the line numbers.

```

1 import numpy as np
2
3 def calculate_energy(mass):
4     """Calculates rest energy E = mc^2"""
5     c = 299792458 # m/s
6     return mass * c**2

```

Listing 2.1: A simple Python script demonstrating the `minted` configuration.

2.5 To-Do Notes

As you draft your thesis, you can leave yourself notes using the `\todo{}` command.

This is an inline todo note, useful for longer comments or reminders inside the text body.

These notes compile into a handy index at the front of the document. When you are ready to finalize your thesis, simply add the `notodo` class option (see [section 1.4](#)) to hide all notes and the list simultaneously.

This is a margin note. It automatically fits the page margin.



First Appendix

This is an example of an appendix chapter. Appendices automatically use alphabetic lettering (Appendix A, Appendix B) instead of standard numeric chapters.

A.1 Supplementary Information

You can include extra derivations, long tables of data, or additional code snippets here that are relevant but too detailed for the main text.